

Semiconductor Package Thermal Parameters Explained

Author: George-Florin Paun
Microchip Technology Inc.

ABSTRACT

This application note provides a comprehensive overview of thermal resistance concepts and calculations for the SOT223-5 package, with MCP1793 Low Drop-out Regulator (LDO) as an example. The objective is to clarify the theory behind thermal resistances, their impact on PCB performance, and practical methods for calculating these resistances in real-world applications. The document addresses essential parameters such as junction-to-ambient thermal resistance (θ_{JA}), junction-to-board thermal resistance (θ_{JB}), and junction-to-top of package characterization parameter (Ψ_{JT}). Through detailed examples and calculations, this document serves as a valuable resource for engineers seeking to optimize thermal performance in electronic designs.

INTRODUCTION

This application note examines the concept of thermal resistance, its usage, and understanding, with a specific focus on the SOT223-5 package. Among other topics, the focus is on the theory behind thermal resistances, their impact on PCB performance, and the methods for calculating thermal resistances in practical applications.

A general purpose DC-DC converter, such as the MCP1793 LDO, with an SOT223-5 lead package serves as a basis for the examples presented in this application note. This package type, commonly used in various electronic applications, offers a convenient and practical example for understanding how thermal resistance impacts the performance and reliability of electronic components. This example will explore the calculation of thermal resistance, identify the key factors that influence it, and apply this knowledge to real-world scenarios to ensure effective thermal management in electronic board designs.

Understanding the thermal resistances of a power Integrated Circuit (IC) is crucial for evaluating heat generation and dissipation. However, effectively managing this heat is essential to ensure long-term performance and reliability, especially in high-power or thermally demanding applications; this is where advanced thermal management techniques become essential. Some of these techniques involve the redistribution or redirection the heat from a specific component to manage its thermal output more efficiently. In electronic design, thermal management solutions may include elements such as heat sinks, thermal vias, or cooling mechanisms that help transfer heat from the component to the surrounding environment or throughout the Printed Circuit Board (PCB).

An example of thermal flow is shown in [Figure 1](#).

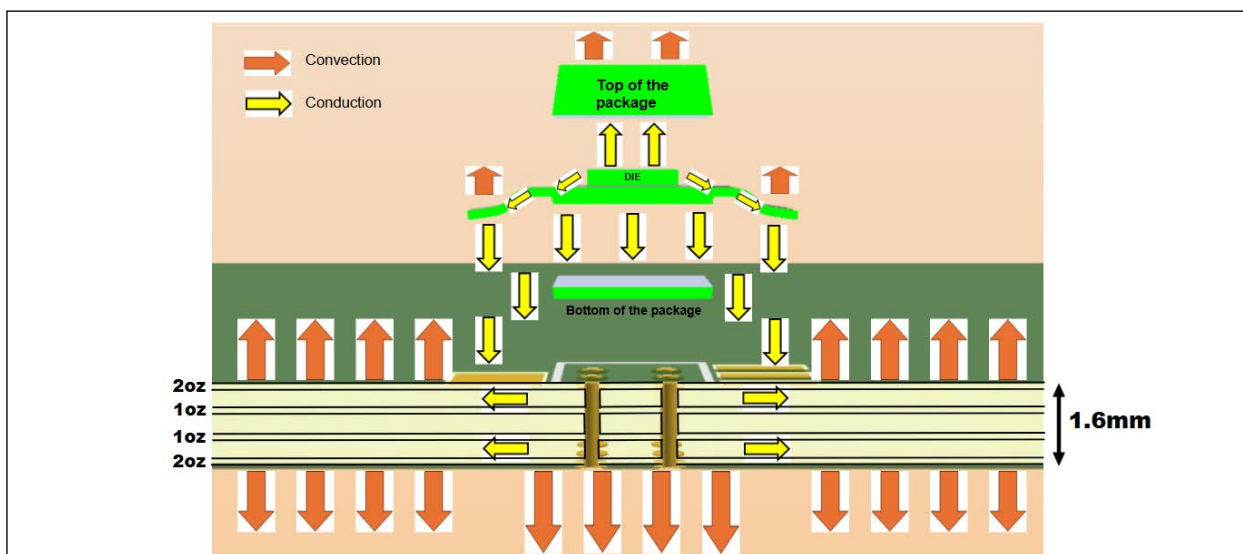


FIGURE 1: Heat Flows through the PCB and the Section Package Case of the SOT223-5 Package.

In summary, these thermal management techniques optimize thermal flow by utilizing materials with different thermal conductivities and strategically positioning thermal management components. This ensures that electronic components remain within the safe operating area temperatures. The process involves both conduction, where heat is transferred directly through materials, and convection, which facilitates the dissipation of heat into the surrounding air or other cooling media, thus helping prevent overheating, improving performance, and increasing the reliability of the electronic system.

For the SOT223-5 package, both conduction and convection significantly contribute to thermal management. The distribution of these roles is the following:

1. **Conduction:** This refers to the transfer of heat through the material of the package itself. When a component is operating, electrical energy is converted into heat, which is conducted from the die (the silicon chip inside the package) through the leads and to the external environment. For the SOT223-5 package, the leads, typically made of metal (such as copper or copper alloy), conduct heat away from the package. The heat flows from the silicon die to the leads and then to the PCB or other surfaces in contact with the leads. Hence, the distribution of heat conduction will occur across the layers where current is intended to flow.

2. **Convection:** This involves the transfer of heat from the surface of the package to the surrounding air. For the SOT223-5 package, the heat conducted through the leads and the package body is released into the surrounding air. As the heat is transferred to the ambient, it causes the air to warm up, and rise, creating natural airflow (natural convection). In cases where airflow is forced (such as fans or blowers), this convective heat transfer is enhanced, helping to dissipate heat more efficiently.

In order to simplify the example of thermal resistance and its representation, an adapted version of thermal resistances can be used, as shown in the detailed view of the package depicted in [Figure 2](#).

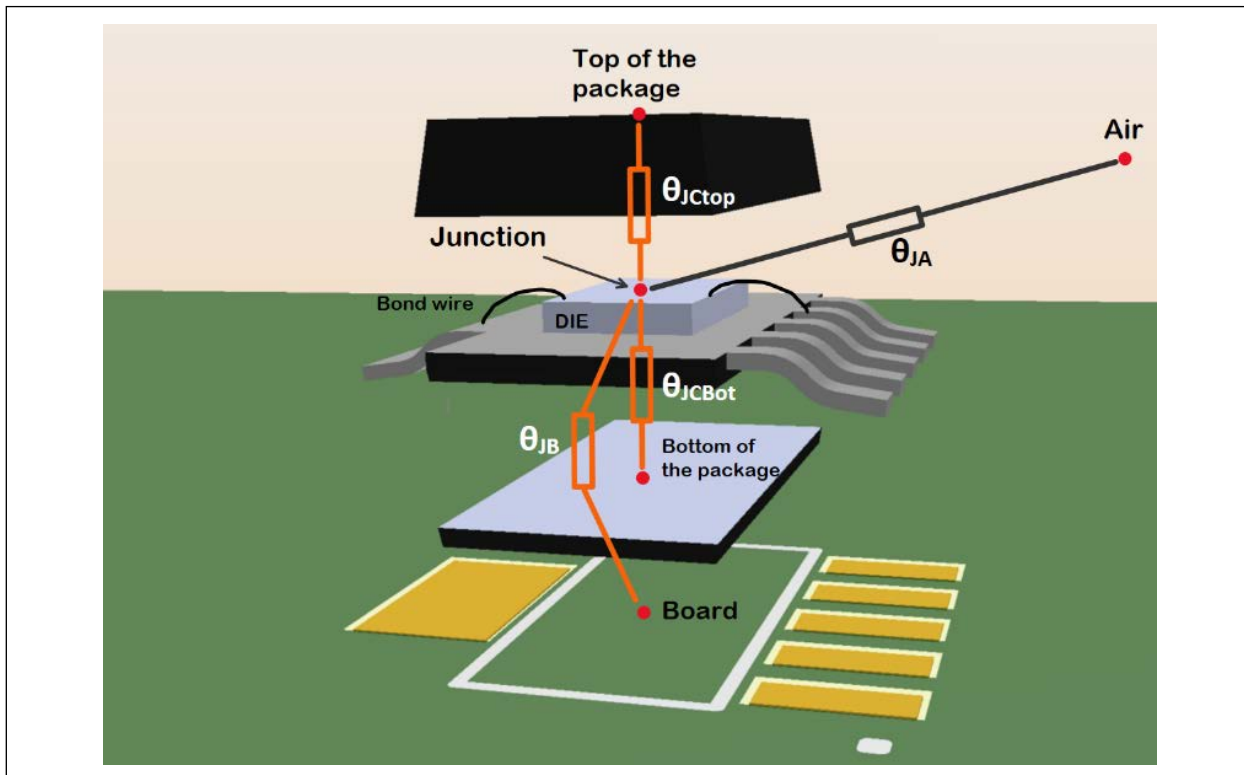


FIGURE 2: Section of the SOT223-5 Package Showing the Thermal Resistances.

Before proceeding, it is important to note that the following parameters provide detailed information in accordance with the JEDEC Thermal Modeling Parameter Terms, as defined in the JESD51-13 standard.

To further understand and gain expertise in the concept of thermal resistance, a calculation example is used, based on the specifications (of the MCP1793 SOT223-5 package) outlined in [Figure 3](#).

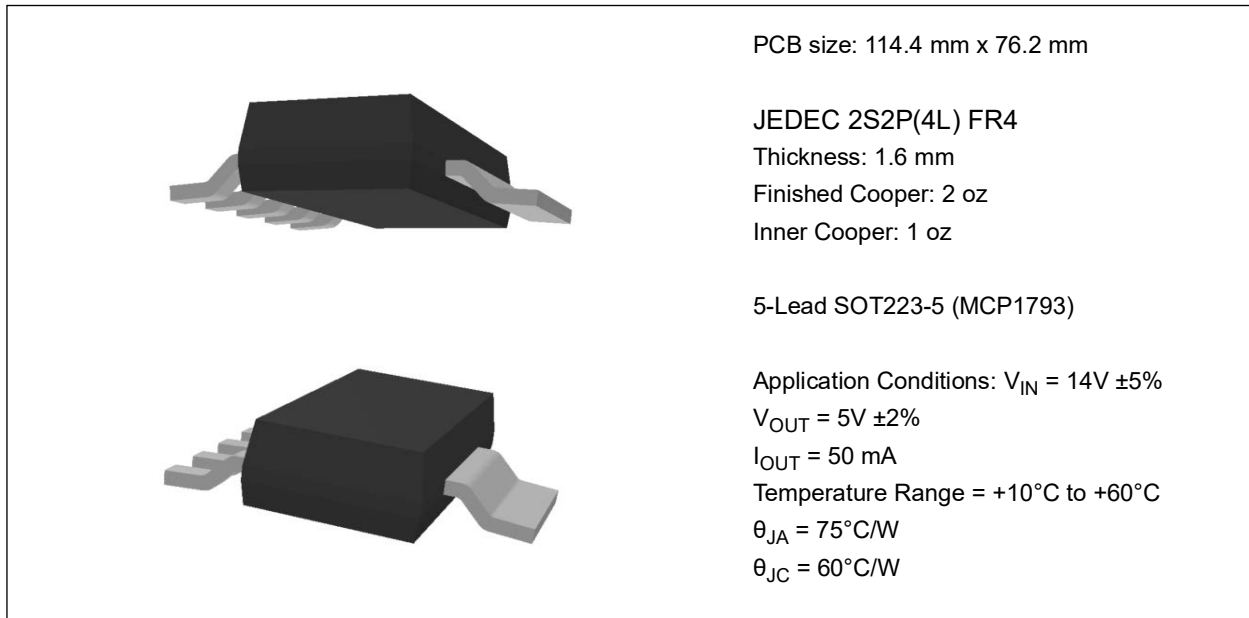


FIGURE 3: Specifications of the Board and SOT223-5 Package.

Considering that the final application PCB size and copper density may differ from the JEDEC standard, thermal parameter estimates should be used only as an initial approximation. Engineering assumptions and margins must also be considered to ensure proper operation.

POWER DISSIPATION

The discussion starts with the internal power dissipation (PD) within the MCP1793, which is dependent on the input voltage, output voltage, output current, and quiescent current. The values used should include the tolerances associated with those parameters and the worst case scenario has to be considered. The internal power dissipation for the Low Dropout Regulator (LDO) can be calculated using the equation below:

$$PD = (V_{IN(MAX)} - V_{OUT(MIN)}) \times I_{OUT(MAX)}$$

Where:

PD	= Internal Power Dissipation
$V_{IN(MAX)}$	= Maximum Input Voltage
$V_{OUT(MIN)}$	= LDO Minimum Output Voltage
$I_{OUT(MAX)}$	= Maximum Output Current

Input Voltage Example

$$V_{IN(MAX)} = 14V + 5\% = 14.7V$$

A +5% tolerance is added to determine the maximum possible input voltage.

LDO Output Voltage and Current Example

$$V_{OUT(MIN)} = 5V - 2\% = 4.9V$$

$$I_{OUT} = 50 \text{ mA}$$

A -2% tolerance is subtracted to determine the minimum possible output voltage.

Power Dissipation Calculation

$$PD = (14.7V - 4.9V) \times 50 \text{ mA}$$

$$PD = 0.49 \text{ Watts}$$

The tolerance added are the worst-case values, used to estimate the maximum internal power dissipation under the most extreme operating conditions. It is a standard practice in electronics design to ensure components are not overstressed or overheated.

It is important to note that the equation may also account for the ground current (I_{GND}). However, if the power dissipation associated with this current is minimal (as is the case for the LDO under consideration), it can be safely neglected.

JUNCTION-TO-AMBIENT THERMAL RESISTANCE (θ_{JA})

Junction-to-Ambient Thermal Resistance (θ_{JA}) (Figure 2) is a key thermal metric that measures the ability of an integrated circuit (IC) package to dissipate heat from the chip to the surrounding environment. It is particularly important under conditions of natural convection. This parameter is defined as the temperature difference between the junction (the hottest part of the chip) and the ambient environment, divided by the power dissipation of the device (PD).

The formula for θ_{JA} is:

$$\theta_{JA} = \frac{T_J - T_A}{PD}$$

Where:

PD = Internal Power Dissipation

The junction temperature (T_J) is calculated based on the ambient temperature (T_A), the power dissipated (PD) and the thermal resistance of the system:

- T_A represents the ambient temperature at the time of measurement.
- T_J is the temperature of the chip's junction while it is operating in an environment with ambient temperature (T_A) and dissipating a power (PD).

Understanding and managing θ_{JA} facilitates designing systems that maintain components within safe operating temperatures, thereby ensuring optimal performance and reliability.

DEVICE JUNCTION TEMPERATURE RISE

The internal junction temperature rise in an electronic component is determined by the internal power dissipation and the thermal resistance from junction-to-ambient (θ_{JA}). This thermal resistance is measured according to EIA/JEDEC standards, namely the JESD51 specification, which outlines the test method and board specifications for measuring θ_{JA} . The actual thermal resistance in a specific application can vary based on various factors such as copper area and thickness.

For more information, refer to Application Note AN792, "A Method to Determine How Much Power a SOT-23 Can Dissipate in an Application" (DS00792).

Calculation Example:

The rise in junction temperature, $T_{J(RISE)}$, can be calculated using the following formula:

$$T_{J(RISE)} = PD \times \theta_{JA}$$

Where:

$T_{J(RISE)}$ = Rise in the Device Junction Temperature over the Ambient Temperature

PD = Internal Power Dissipation (calculated at worst case condition)

θ_{JA} = Thermal resistance from Junction-to-Ambient (75°C/W for SOT-223 package)

Given the previously obtained values for $PD_{(MAX)}$ and θ_{JA} :

$$T_{J(RISE)} = 0.49W \times 75^\circ C/W$$

$$T_{J(RISE)} = 36.75^\circ C$$

This means the junction temperature will rise by 36.75°C above the ambient temperature due to power dissipation.

JUNCTION TEMPERATURE ESTIMATE

To estimate the internal junction temperature, the calculated temperature rise is added to the ambient or off-set temperature. For this example, the worst-case junction temperature is estimated below.

To find the junction temperature (T_J), apply:

$$T_J = T_{J(RISE)} + T_A$$

Where:

T_J = Junction Temperature

$T_{J(RISE)}$ = Rise in the Device Junction Temperature over the Ambient Temperature

T_A = Ambient Temperature (maximum temperature in the operating range of the application)

$$\begin{aligned} T_J &= T_{J(RISE)} + T_A \\ T_J &= 36.75^\circ C + 60.0^\circ C \\ T_J &= 96.75^\circ C \end{aligned}$$

Subsequently, the maximum power dissipation of the device can be calculated if the θ_{JA} and maximum allowed junction temperature rise are known.

JUNCTION-TO-CASE (θ_{JC})

θ_{JC} specifies the thermal resistance from the junction to the case surface. Some manufacturers may specify the surface at which the measurement is taken, providing thermal data as θ_{JCtop} and θ_{JCbot} . However, in Microchip data sheets, θ_{JC} is typically provided as a single value. The reference point for measuring the case temperature is the hottest point on the package, which is usually the center of the package surface or the device lid.

θ_{JC} is particularly useful to estimate the thermal performance of a package when a heat-sink is attached. Depending on the use-case of the package, if only one θ_{JC} is mentioned, it refers to the surface used for heat extractions. In particular cases where both a heat-sink and the bottom exposed pad is used for heat dissipation, to avoid confusions both θ_{JCtop} and θ_{JCbot} are used.

JUNCTION-TO-BOARD THERMAL RESISTANCE (θ_{JB})

Junction-to-Board Thermal Resistance (θ_{JB}) represents the temperature difference between the junction (IC die) and the PCB divided by the power dissipation of the IC. The heat transfer between the die and the board is accounted for by considering multiple paths such as junction-to-pin-to-board and junction-through-plastic-to-board. Although θ_{JB} assumes a uniform board temperature, which is not entirely accurate, it provides a useful approximation for the initial thermal modeling.

Formula:

$$\theta_{JB} = \frac{T_J - T_B}{PD}$$

Where:

T_J = Junction Temperature
 T_B = Board Temperature
 PD = Internal Power Dissipation

SEMICONDUCTOR PACKAGE THERMAL CHARACTERIZATION PARAMETERS

Compared to the previous thermal resistances, a characterization parameter is an empirical parameter, which compared to the resistance which assumes only one main path of heat flow, assumes that the heat flows through all available paths.

Junction-to-Top of Package Characterization Parameter (Ψ_{JT})

Ψ_{JT} indicates the thermal behavior between the active junction of the device and the top center of the package, with respect to how heat is transferred. This metric helps estimate junction temperatures based on measured case temperatures and is defined in JESD51-2.

$$T_J = T_C + \Psi_{JT} \times PD$$

Where:

T_J = Junction Temperature
 T_C = Top Case Temperature
 PD = Internal Power Dissipation.

Junction-to-Board Characterization Parameter (Ψ_{JB})

Ψ_{JB} is similar to Ψ_{JT} and is defined by JESD51-6. It depicts the thermal behavior between the active junction of the device and the center package pin temperature (e.g., the exposed pad on the PCB).

Formula:

$$T_J = T_{(PCB)} + \Psi_{JB} \times PD$$

Where:

T_J = Junction Temperature
 $T_{(PCB)}$ = Temperature of the PCB
 PD = Internal Power Dissipation

It is important to note that Ψ_{JB} is similar to $R\theta_{JB}$, as the PCB dissipates 75% to 95% of the device's heat.

Conclusions:

Understanding and accurately measuring thermal resistance parameters is crucial for evaluating the thermal performance of electronic components. Each parameter, such as θ_{JA} , θ_{JB} , Ψ_{JT} , and Ψ_{JB} , provides unique insights into how heat is managed within a device and its surrounding environment. Effectively applying these metrics helps engineers design effective cooling solutions, thereby ensuring the reliability and longevity of electronic components.

By considering these thermal metrics and their appropriate applications, engineers can optimize the thermal management of electronic devices, leading to improved performance and extended component lifespan. Accurate thermal modeling and the correct use of thermal resistance parameters are essential for preventing overheating and ensuring the reliable operation of electronic components in various environmental conditions.

REFERENCES

1. *Electronic Industries Alliance JEDEC Solid State Technology Association EIA/JEDEC Standard, High Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages, JESD51-7, February 1999.*
2. *Understanding the JEDEC Integrated Circuit Thermal Test Standards, QPEDIA JESD51-6 Advanced Thermal Solutions, Inc.*
3. *JEDEC Solid State Technology Association, JEDEC Standard Test Boards for Area Array Surface Mount Package Thermal Measurements, JESD51-9, July 2000.*
4. *Micrel's Guide to Designing with Low-Dropout Voltage Regulators, Revised Edition, December 1998.*
5. [The Junction-to-Case Thermal Resistance: A One-Dimensional Underachiever in a Three-dimensional, Conjugate Heat Transfer World](#), *Electronics Cooling*, April 2018.
6. [MCP1792/3 Data Sheet](#), "100 mA High-Voltage Automotive LDO", Microchip Technology Inc., 2022, (DS20006229).

Microchip Information

Trademarks

The “Microchip” name and logo, the “M” logo, and other names, logos, and brands are registered and unregistered trademarks of Microchip Technology Incorporated or its affiliates and/or subsidiaries in the United States and/or other countries (“Microchip Trademarks”). Information regarding Microchip Trademarks can be found at <https://www.microchip.com/en-us/about/legalinformation/microchip-trademarks>.

ISBN: 979-8-3371-1499-6

Legal Notice

This publication and the information herein may be used only with Microchip products, including to design, test, and integrate Microchip products with your application. Use of this information in any other manner violates these terms. Information regarding device applications is provided only for your convenience and may be superseded by updates. It is your responsibility to ensure that your application meets with your specifications. Contact your local Microchip sales office for additional support or, obtain additional support at www.microchip.com/en-us/support/design-help/client-support-services.

THIS INFORMATION IS PROVIDED BY MICROCHIP "AS IS". MICROCHIP MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WHETHER EXPRESS OR IMPLIED, WRITTEN OR ORAL, STATUTORY OR OTHERWISE, RELATED TO THE INFORMATION INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY, AND FITNESS FOR A PARTICULAR PURPOSE, OR WARRANTIES RELATED TO ITS CONDITION, QUALITY, OR PERFORMANCE.

IN NO EVENT WILL MICROCHIP BE LIABLE FOR ANY INDIRECT, SPECIAL, PUNITIVE, INCIDENTAL, OR CONSEQUENTIAL LOSS, DAMAGE, COST, OR EXPENSE OF ANY KIND WHATSOEVER RELATED TO THE INFORMATION OR ITS USE, HOWEVER CAUSED, EVEN IF MICROCHIP HAS BEEN ADVISED OF THE POSSIBILITY OR THE DAMAGES ARE FORESEEABLE. TO THE FULLEST EXTENT ALLOWED BY LAW, MICROCHIP'S TOTAL LIABILITY ON ALL CLAIMS IN ANY WAY RELATED TO THE INFORMATION OR ITS USE WILL NOT EXCEED THE AMOUNT OF FEES, IF ANY, THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THE INFORMATION.

Use of Microchip devices in life support and/or safety applications is entirely at the buyer's risk, and the buyer agrees to defend, indemnify and hold harmless Microchip from any and all damages, claims, suits, or expenses resulting from such use. No licenses are conveyed, implicitly or otherwise, under any Microchip intellectual property rights unless otherwise stated.

Microchip Devices Code Protection Feature

Note the following details of the code protection feature on Microchip products:

- Microchip products meet the specifications contained in their particular Microchip Data Sheet.
- Microchip believes that its family of products is secure when used in the intended manner, within operating specifications, and under normal conditions.
- Microchip values and aggressively protects its intellectual property rights. Attempts to breach the code protection features of Microchip product is strictly prohibited and may violate the Digital Millennium Copyright Act.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of its code. Code protection does not mean that we are guaranteeing the product is “unbreakable”. Code protection is constantly evolving. Microchip is committed to continuously improving the code protection features of our products.